

QC SKILL · RECONSTRUCTION SPECIFICATION

qc-coherence

Compositional consistency and enforcement of the invariant registry. Static contracts versus observed composition. Requires qc-core.

Canonical source of truth is the **SKILL.md** tree in this repository. This PDF reconstructs that tree for porting. Do not treat this PDF as a second design. Do not vendor a specialization without **qc-core**.

Role

REQUIRED: load qc-core first. Laws, ledger schema 1.1, verdict, .qc-profile.json, Discovery+Verify, deferred vocabulary, CLEAN lists, and clustering artifacts live there. This skill does not reimplement them.

Guarantee **compositional consistency**: every public contract has complete and behaviorally aligned implementations; domain concepts, errors, and data shapes use single canonical forms; architectural boundaries and cross-cutting operational concerns are uniform; previously discovered invariants remain enforced. Produces maps and an invariant registry, not one-shot opinions.

C1, MS-4, and A1 exist to catch divergence between **static contracts (system as imagined)** and **observed composition (system as found)**. Hardening owns the runtime proof of sequences and live adapters. This skill owns static contracts and canonical forms. Durable artifacts (invariants[], canonical_forms[], layer_model, clusters[]) are the bridge.

Four sub-audits (dependency order)

#	Sub-audit	Yield
1	Structural (S1–S5)	Deterministic. Import graph feeds partitioning. Mechanical hits skip dual-verify.
2	Conformance (C1)	Protocol/implementation matrix. Behavioral and data-contract parity across backends.
3	Semantic (M1–M5)	Naming, error taxonomy, serialization. Accepted clusters write canonical_forms[].
4	Architectural (A1–A5)	Dependency direction, layers, invariants. A1 writes layer_model. A5 is operational consistency in place.

Judgment-heavy checks (M1, A3, MS-2) **MUST** write clusters[] and the corresponding durable map. Later runs load these as ground truth and only re-flag regressions. Semantic/Architectural findings use the scenario shape: under input sequence X, backend A returns Y while B returns Z.

Module scope (--scope=module)

Six composition checks (formerly a separate structural skill). Accepted debt lives in profile deferred[], not a sidecar markdown file.

Id	Check
MS-1	Protocol conformance parity
MS-2	Responsibility density (writes clusters[])
MS-3	Coordination surface
MS-4	Backend parity — when 2+ backends exist, run immediately after MS-1; not skippable if pass_debt/hot_spots name adapters
MS-5	Cross-session consistency

Id	Check
MS-6	Test architecture

Workflow

Sub-audits 2–4 use `qc-core workflows/discovery-verify.js` with `kind: "lenses"`. There is no per-skill `coherence-discovery-verify.js`. Both votes must say `real → fix`; otherwise `defer`. CLEAN lists are named `{artifact, invariant}`.

Prior-skill inputs

Load `.qc-profile.json` and the hardening ledger `.qc-findings/qc-hardening.json`. Deferred hardening findings are `priorState` by default; link `related_findings` instead of minting a parallel id. If a new implementation or module was added, `partition.py --reexam` neighbors that are existing backends or hot spots get C1/MS-4 behavioral parity even if they did not change.

P0 policy

Open P0/P1 for this skill. Suite human-gate still blocks on any P0. Prose COHERENT / DRIFTING / ACTION REQUIRED maps onto READY / READY_WITH_DEBT / NOT_READY via `qc-core`.

A run is incomplete if findings were handled but `canonical_forms[]` / `layer_model` / `invariants[]` were not updated. Empty findings plus updated artifacts is a valid win.